FTS-Diffusion Replication Attempt: Results and Diagnostic Report

May 11, 2026

1 What We Did

- Collected the paper-consistent artifacts: 7 TATR matrices, 7 final figures under `figures/`, reference TMTR/TATR summaries, SISC outputs, and bundled reference figures.
- Aggregated each TATR matrix over 30 runs and plotted mean MAE with one-standard-deviation bands.
- Added the predictable-drift component as a separate diagnostic, but did not report FTS-Diffusion drift numbers because no ordered generated sample paths are present in this checkout.

2 Available Artifacts

The report uses the paper-consistent K choices: GOOG K=11, ZCF K=11, and SP500 K=14. Other local artifacts, such as ZCF K=14, are excluded from the tables and figures for consistency. The TATR matrices have columns 0, 10, ..., 100, interpreted as synthetic augmentation percentages. The full file list and plot list are stored in `manifest.json` and `tables/tatr_summary.csv`. No real generated-sample drift diagnostic outputs are present in this checkout; this report therefore does not claim no-drift results for FTS-Diffusion samples.

3 TATR Matrix Results

Table 1 summarizes the pulled TATR matrices. Negative change means lower MAE than the 0% synthetic baseline.

Table 1: Summary of pulled TATR result matrices. Negative change means lower MAE than the 0% synthetic baseline.

Data	K	Variant	Runs	0% mean	Best %	Best mean	Best change %	100% mean	100% change %
GOOG	K11	authors	30	0.0119	0	0.0119	0.0000	0.1308	1003.0943
GOOG	K11	single	30	0.0119	0	0.0119	0.0000	0.0244	105.6372
SP500	K14	authors	30	0.0611	0	0.0611	0.0000	0.1886	208.3886
SP500	K14	random_init	30	0.0611	0	0.0611	0.0000	0.1028	68.1659
SP500	K14	single	30	0.0611	50	0.0132	-78.4489	0.0173	-71.6390
ZCF	K11	authors	30	0.0113	0	0.0113	0.0000	0.0198	75.0735
ZCF	K11	single	30	0.0113	0	0.0113	0.0000	0.0600	430.8901

4 Updated TATR/TMTR Result Figures

These are the newest final result PDFs under the repository `figures` directory, filtered to GOOG K=11, ZCF K=11, and SP500 K=14.

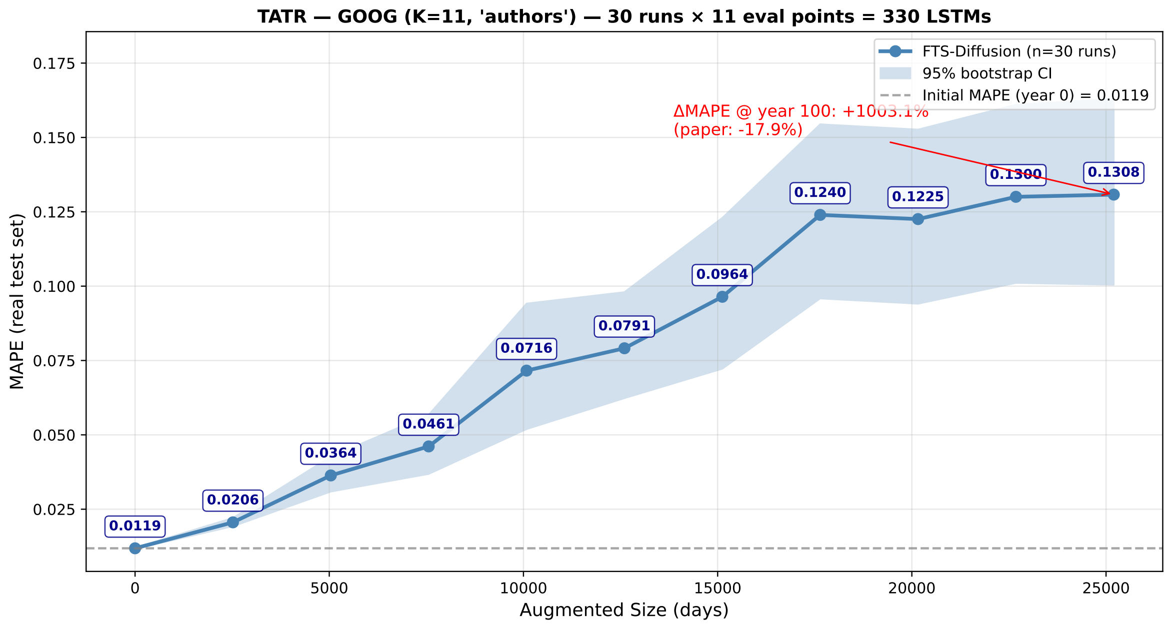

Figure 1: Pulled final result figure for GOOG K11 authors.

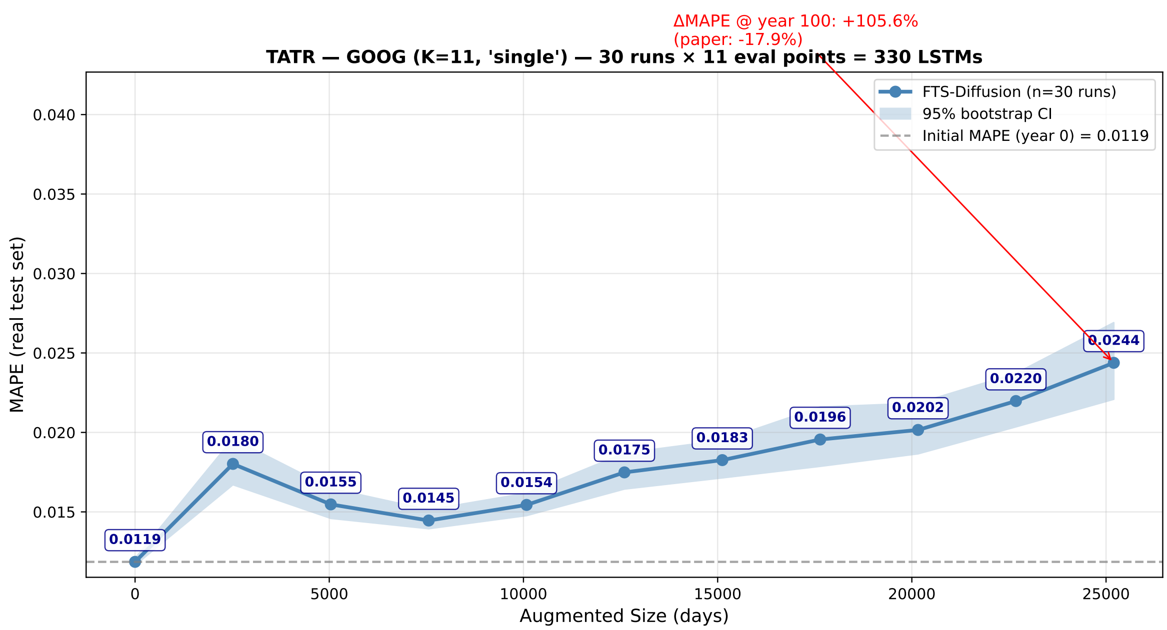

Figure 2: Pulled final result figure for GOOG K11 single.

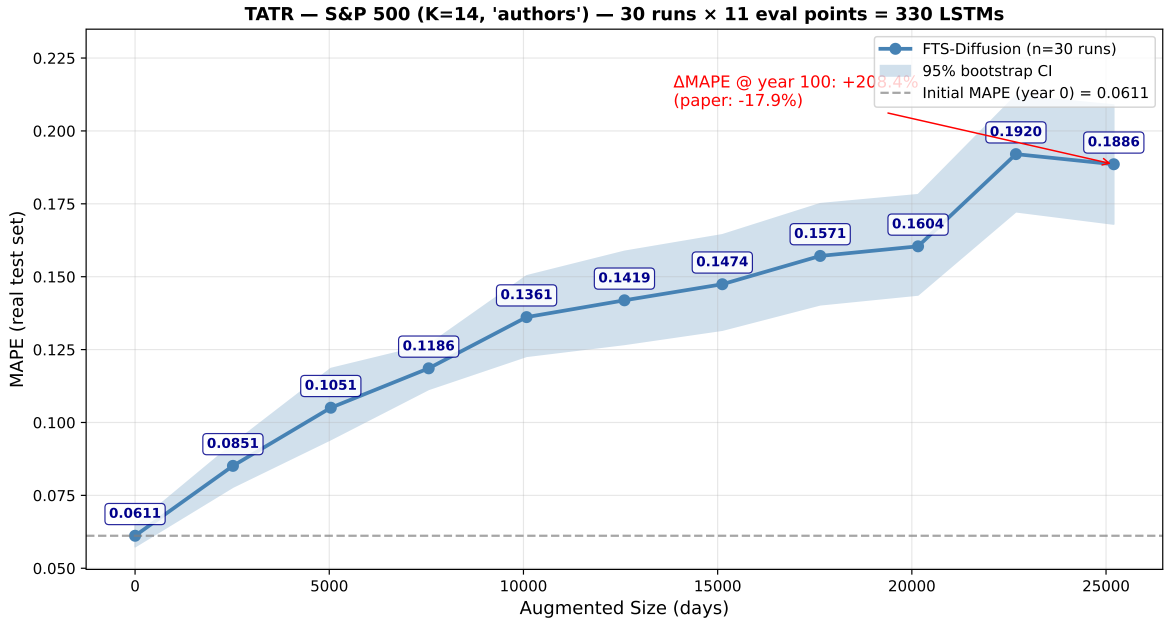

Figure 3: Pulled final result figure for SP500 K14 authors.

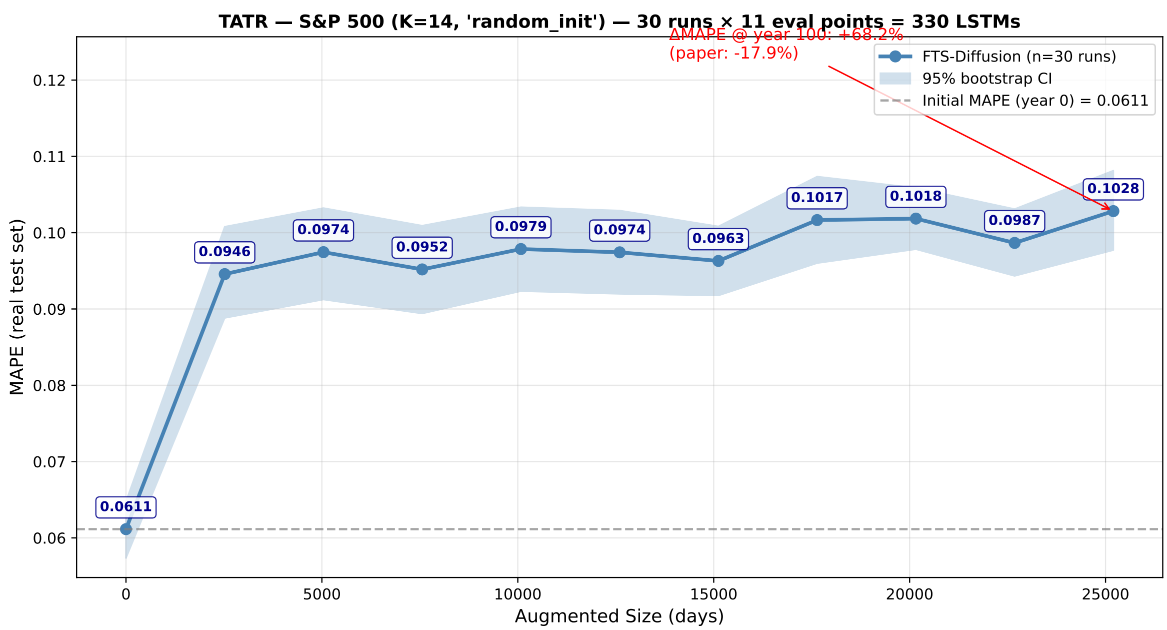

Figure 4: Pulled final result figure for SP500 K14 random_init.

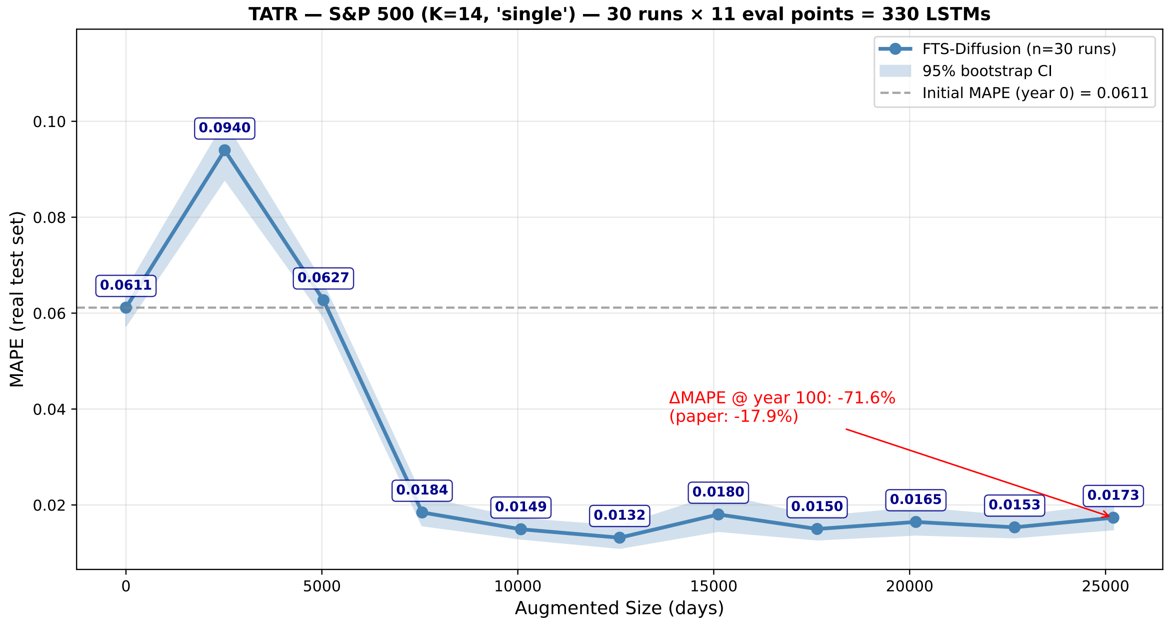

Figure 5: Pulled final result figure for SP500 K14 single.

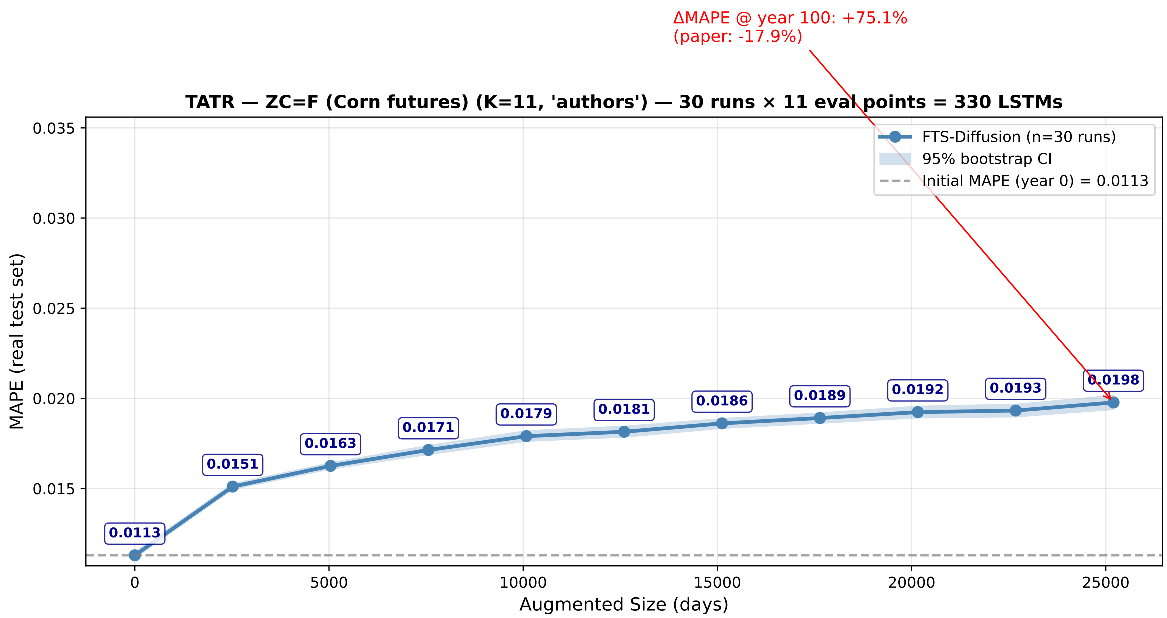

Figure 6: Pulled final result figure for ZCF K11 authors.

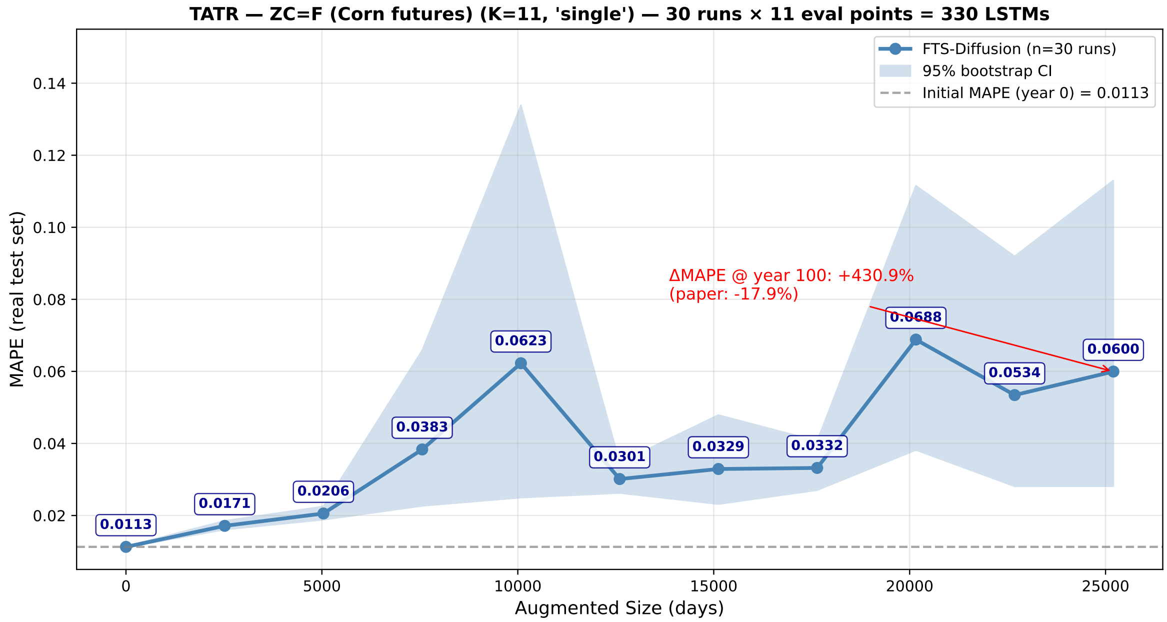

Figure 7: Pulled final result figure for ZCF K11 single.

5 Reference Summary CSVs

Table 2 reproduces the bundled reference TMTR/TATR summary rows.

Table 2: Reference summary CSV rows bundled with the original FTS-Diffusion repository.

Exp.	Idx	Avg
TATR	0	0.1509
TATR	1	0.1254
TATR	2	0.2129
TATR	3	0.2305
TATR	4	0.2004
TATR	5	0.2234
TATR	6	0.2071
TATR	7	0.2710
TATR	8	0.3173
TATR	9	0.2781
TATR	10	0.3084
TMTR	0	0.2959
TMTR	1	0.3117
TMTR	2	0.3299
TMTR	3	0.4419
TMTR	4	0.4973

6 SISC Pattern-Recognition Artifacts and Plots

The local reference artifacts contain 14 SISC centroids and 704 segmentation boundary rows for SP500 K=14.

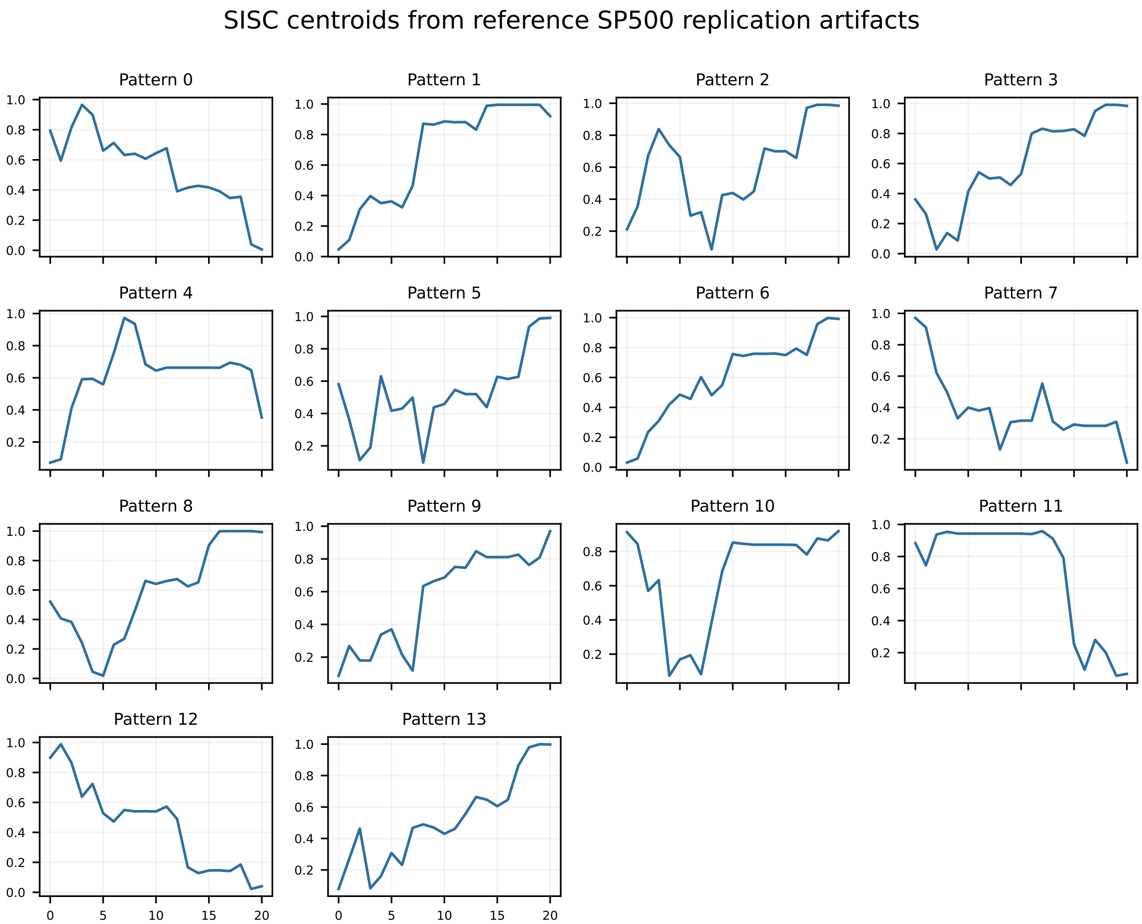

Figure 8: SISC artifact plot: sisc centroids.

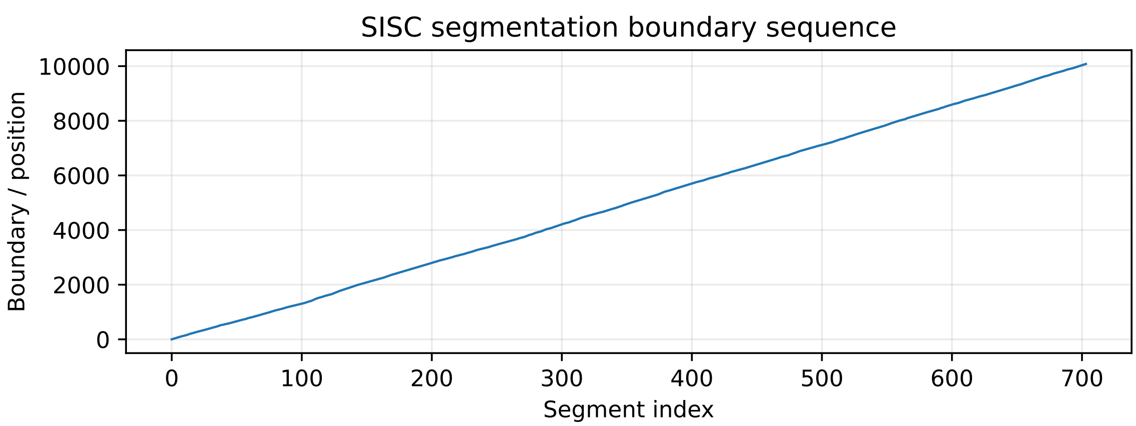

Figure 9: SISC artifact plot: sisc segmentation boundaries.

7 Bundled Stylized-Fact Figure

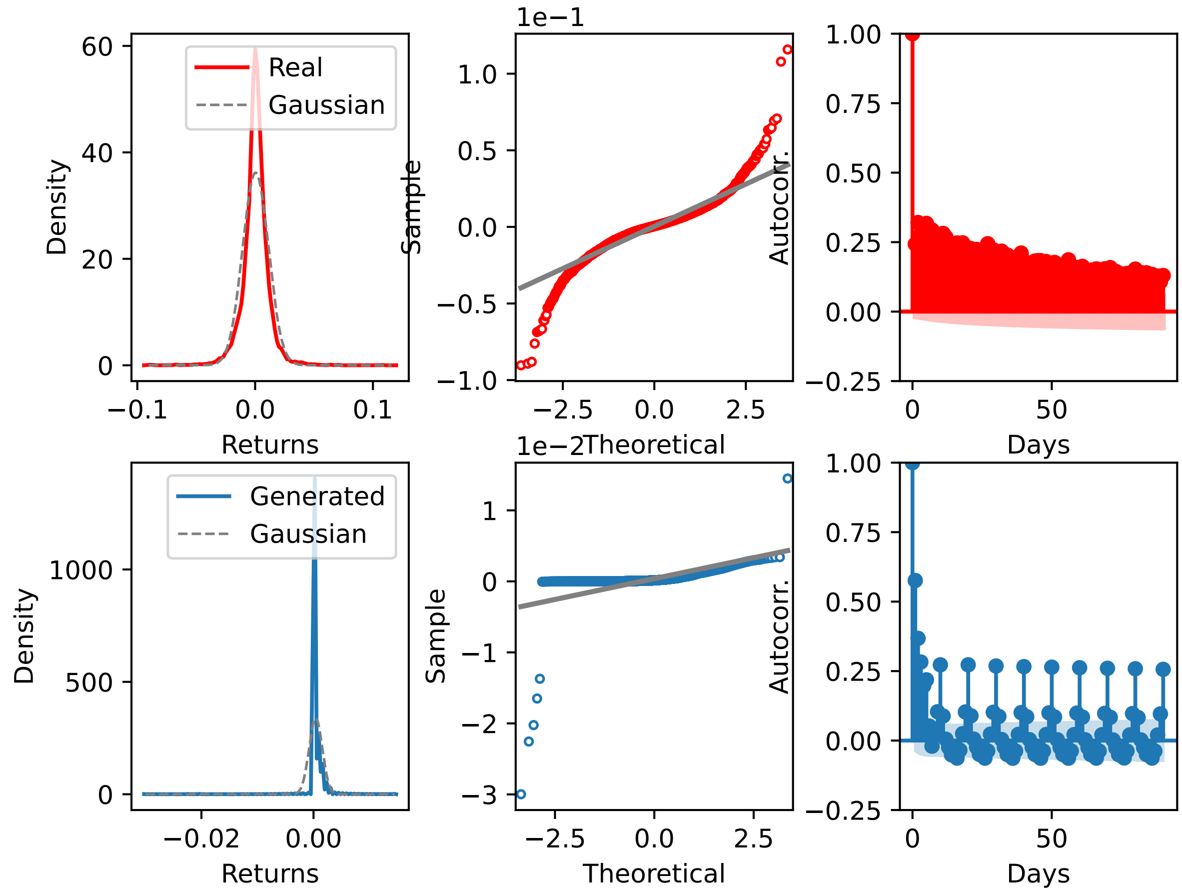

Figure 10: Reference stylized-fact figure bundled with the FTS-Diffusion code.

8 Predictable-Drift Extension Status

The drift component targets the gap that matching stylized facts does not imply

$$\mathbb{E}[r_{t+1} \mid \mathcal{F}_t] = 0.$$

It fixes

$$\omega_t = (1, r_t, r_{t-1}, |r_t|, |r_{t-1}|)$$

and computes

$$\widehat{\delta} = \left\| \frac{1}{T} \sum_t r_{t+1}\omega_t \right\|_2 .$$

This is a finite-dimensional check, not a full no-arbitrage proof. Because no ordered generated return paths are present here, this PDF does not report drift statistics for actual FTS-Diffusion samples.

9 Replication Caveats

- The pulled `results/tatr/**/results_matrix.csv` files are result matrices, not generated financial return paths.

- The report does not include RCGAN, TimeGAN, or CSDI reruns.
- The SISC artifacts and reference figures are bundled outputs from the reference directory; they support comparison, but are not proof that the full pipeline was freshly retrained in this checkout.
- A full replication report should add training logs, generated sample CSVs, exact seeds, and final drift diagnostics once those outputs are available.